

Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs

Anonymous Author(s)

Abstract

Intervention harvesting [1] estimates the position-bias ratio $p_k/p_{k'}$ by aggregating per-cell click-rate ratios across many (query, doc) cells. On real click logs the per-cell impression counts are extremely heavy-tailed: on the full KDD Cup 2012 Track 2 log, 60% of interventional (q, ad) pairs have ≤ 5 impressions while 0.1% of pairs hold 42% of all impressions. The way cells are weighted is therefore decisive. The *harmonic* weight, used in the intervention-harvesting literature, equals the inverse-variance choice under the Position-Based Model (PBM) once cell-level relevance is dropped from the per-cell variance—a *relevance-agnostic surrogate* of the actual variance—and a Cauchy–Schwarz inequality guarantees harmonic reduces this surrogate for any impression counts. The actual inverse-variance solution requires per-cell relevance and is infeasible at this scale. We further argue that the naive pair bootstrap is poorly aligned with the conditional-on-counts efficiency question on heavy-tailed data, and adopt instead the textbook delta-method asymptotic variance of the weighted ratio estimator [12, 14] as a closed-form diagnostic. On the full KDD log harmonic’s asymptotic relative efficiency over uniform weighting is 4.5×, matched within 1–2% by click-resample standard error and robust to weight capping. A Cochran–Q test confirms substantial PBM mis-specification; under such mis-specification, different weight schemes target different weighted averages of cell-specific ratios—an estimand caveat we make explicit. Synthetic and Yahoo-LTR semi-synthetic experiments with ground truth corroborate the efficiency prediction (up to 54% variance reduction at high imbalance).

ACM Reference Format:

Anonymous Author(s). 2026. Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs. In *Proceedings of ACM CIKM '26*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [9]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions (1, 2)), **60.7% of**

pairs have $N_1 + N_2 \leq 5$ while 0.1% of pairs carry 42% of all impressions [10]. Public benchmarks can hide this shape: the released Baidu-ULTR [17] aggregation has 94% of (q, d, pos) cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$, which we attribute to the per-cell deduplication used in the public release rather than to the underlying production logs (the unaggregated KDD release, by contrast, preserves the heavy-tailed production distribution).

Equal-weighting heteroscedastic estimators is statistically sub-optimal by a century of classical theory: Aitken’s generalised-least-squares theorem [2] mandates inverse-variance weights, a result the meta-analysis literature [5, 8] has championed for fifty years. Recent ULTR work [1, 15] introduced *harmonic* weighting $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ to reduce variance versus uniform on heavy-tailed counts, but existing validation has relied on *bootstrap* resampling, which we show can be misleading for inverse-variance-style weights on singleton-dominated heavy-tailed data (Section 3).

Contributions. (1) We identify harmonic as the inverse-variance optimum on a *relevance-agnostic count-only surrogate* of the delta-method variance under PBM (Theorem 2); prove an unconditional Cauchy–Schwarz reduction of $V_k + V_{k'}$ that is conditional on the surrogate objective and unconditional in the counts (Theorem 3); and characterise the exact asymmetric surrogate oracle of which harmonic is the $\alpha = \beta$ special case (Theorem 4). (2) We adopt the plug-in delta-method asymptotic variance $\tilde{V}(\omega)$ as a finite-sample efficiency diagnostic aligned with the conditional-on-counts question, and use empirical click-resample SE as its calibration baseline; we discuss why pair-bootstrap is misaligned with this question on heavy-tailed data. (3) On the full KDD log, ARE(harmonic, uniform) = 4.53 on the surrogate, calibrated within 1–2% and robust to weight-capping; PBM mis-specification (Cochran–Q/df = 3.83) makes ω a precision/estimand trade-off, not a search for a single “true” $p_k/p_{k'}$. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Under PBM, $P(\text{click} \mid q, d, k) = p_k r_{q,d}$. Given an interventional set $I_{k,k'}$ of (q, d) pairs that appeared at both positions k and k' , the position-bias ratio is estimated as [1]

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \hat{\theta}_k^i}{\sum_i \omega_i \hat{\theta}_{k'}^i}, \quad \hat{\theta}_k^i = \frac{C_k^i}{N_k^i}. \quad (1)$$

This is a self-normalised weighted ratio estimator. We say a weight scheme is *ratio-unbiased* when the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$, i.e. $\mathbb{E}[A \mid N]/\mathbb{E}[B \mid N] = p_k/p_{k'}$ (not $\mathbb{E}[\hat{R}] = p_k/p_{k'}$; the finite-sample bias of the self-normalised ratio is $O(1/N)$ under standard regularity). Under PBM, the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$ for any count-only weight $\omega_i = \omega(N_k^i, N_{k'}^i)$, and

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CIKM '26, Toronto, Canada

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

the resulting self-normalised ratio estimator is consistent for this target [1].

PROPOSITION 1 (CONDITIONAL DELTA-METHOD VARIANCE). *By the delta method [14], $\text{Var}(\hat{R} \mid \mathbf{N}) \approx R^2[\alpha V_k(\omega) + \beta V_{k'}(\omega)]$ with $V_r(\omega) = \sum_i \omega_i^2 / N_r^i / (\sum_i \omega_i)^2$, $\alpha = (1 - p_k)/p_k$, $\beta = (1 - p_{k'})/p_{k'}$.*

Relevance-agnostic count-only IVW. The exact Bernoulli variance is $\text{Var}(\hat{\theta}_k^i) = p_k r_i (1 - p_k r_i) / N_k^i$, which depends on the cell relevance r_i . Plug-in inverse-variance weights with $\hat{r}_i$ estimated per cell define the θ -aware oracle of Theorem 4; in heavy-tailed click logs $\hat{r}_i$ is itself extremely noisy on the $\sim 60\%$ singleton mass, making this oracle infeasible in practice (Table 2: ARE = 0.002). We therefore work with a *relevance-agnostic surrogate* that drops the r_i -dependent factor from the cell-level variance. The inverse-variance optimum of Aitken [2] on this surrogate, restricted to count-only weights, is the harmonic mean of the two cell counts. We make this precise:

THEOREM 2 (COUNT-ONLY IVW ON THE SURROGATE). *Among all non-negative weights $\omega_i = \omega(N_k^i, N_{k'}^i)$, harmonic weighting minimises the count-only delta-method variance surrogate $V_k(\omega) + V_{k'}(\omega)$.*

THEOREM 3 (SURROGATE GUARANTEE, UNCONDITIONAL IN COUNTS). *For any impression counts $\{N_k^i, N_{k'}^i\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(1) + V_{k'}(1)$. The guarantee is unconditional in the counts but conditional on $V_k + V_{k'}$ as the objective; it implies improved $\text{Var}(\hat{R} \mid \mathbf{N})$ via Proposition 1 when $\alpha \approx \beta$. For the asymmetric $\alpha V_k + \beta V_{k'}$ surrogate the bound is empirical only, but we verify on full KDD that harmonic reduces it by essentially the same factor as the symmetric one (Section 4).*

SKETCH. Substituting $\omega_i^h = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ collapses the LHS to $1/W$ with $W = \sum_i \omega_i^h$. The RHS is $(1/n^2) \sum_i 1/\omega_i^h$. Cauchy-Schwarz gives $n^2 \leq W \sum_i 1/\omega_i^h$. $\square$

THEOREM 4 (EXACT ASYMMETRIC ORACLE). $\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq (\sum_i 1/s_i)^{-1}$ with $s_i = \alpha/N_k^i + \beta/N_{k'}^i$, attained at $\omega_i^* \propto N_k^i N_{k'}^i / (\alpha N_k^i + \beta N_{k'}^i)$. Harmonic is the $\alpha = \beta$ special case.

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count heuristic of [1], which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true α, β into ω^* yields variance within $\pm 2\%$ of harmonic. Complete proofs and additional sensitivity analyses are provided in the supplementary material.

Limits of the symmetric guarantee. The unconditional bound in Theorem 3 concerns $V_k + V_{k'}$. The asymmetric analogue $\alpha V_k(\omega^h) + \beta V_{k'}(\omega^h) \leq \alpha V_k(1) + \beta V_{k'}(1)$ does *not* hold unconditionally: a closed-form $n = 2$ counterexample with $(N_k^1, N_{k'}^1) = (1, 100)$, $(N_k^2, N_{k'}^2) = (100, 1)$ and $(\alpha, \beta) = (0.01, 1)$ has harmonic strictly worse than uniform on the asymmetric surrogate. The configurations under which this fails are degenerate (each cell informative at exactly one position); we have not encountered them in any realistic count distribution we tested. The ω^* in Theorem 4 remains the exact oracle for all (α, β) .

3 Plug-in Asymptotic Variance: the Right Instrument

$\tilde{V}$ in closed form. With $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at k and k' conditionally independent given counts, the delta method

Table 1: Plug-in SE vs. click-resample SE across KDD subsamples, and ARE(harmonic, uniform) at each scale. Click resample only computed at small f for runtime; agreement is within 1–2% where measured. ARE grows with data volume because heavy-tail mass becomes more pronounced at larger scales.

f	pairs	$\widehat{\text{SE}}_U^{\text{plug}}$	$\widehat{\text{SE}}_U^{\text{boot}}$	$\widehat{\text{SE}}_H^{\text{plug}}$	ARE
10^{-3}	4.3K	0.131	0.132	0.111	1.66
10^{-2}	54K	0.039	0.038	0.028	2.21
10^{-1}	542K	0.012	–	0.007	3.07
1.0	4.61M	0.0042	–	0.0020	4.53

gives

$$\tilde{V}(\omega) = \frac{\sum_i \omega_i^2 \theta_k^i (1 - \theta_k^i) / N_k^i}{A^2} + \frac{\sum_i \omega_i^2 \theta_{k'}^i (1 - \theta_{k'}^i) / N_{k'}^i}{B^2}, \quad (2)$$

$\text{SE}(\hat{R}) = |\hat{R}| \sqrt{\tilde{V}(\omega)}$. Plugging in $\hat{\theta}_i$ (with the shrinkage estimator $\hat{\theta}_i = (C_i + \bar{\theta}_k) / (N_i + 1)$, $\bar{\theta}_k$ the marginal position- k CTR) yields the textbook self-normalised IS variance estimator [11–13].

ARE. For schemes a, b , $\text{ARE}(a, b) = \tilde{V}(\omega_b) / \tilde{V}(\omega_a)$: scheme b needs $\text{ARE}(a, b)$ times the data of a for equal precision. $\tilde{V}$ scales as $1/\text{ESS}(\omega)$, with Kong’s effective sample size [11] $\text{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$.

Why pair-bootstrap is misaligned here. A naive bootstrap that re-samples (q, d) pairs estimates a different quantity than the conditional-on-counts efficiency we care about, for three reasons. (i) Averaging over $\sim 60\%$ near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects averaging-of-noise rather than estimator precision. (ii) Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. (iii) Pair-bootstrap entangles estimator efficiency with model fit; $\tilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample (parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

Calibration and stability at scale. Table 1 reports plug-in SE versus empirical click-resample SE across KDD sample fractions, together with ARE(harmonic, uniform) at each scale. Because both $\tilde{V}$ and click-resample SE share the conditional Bernoulli + cross-cell-independence assumption, their agreement is primarily an implementation-correctness check; a stronger probe of the independence assumption would be a query-cluster bootstrap, which is sensitive to Cochran- Q -style heterogeneity. We report pair-bootstrap numbers in supplementary material. The ARE grows with sample size ($1.66 \rightarrow 4.53$) because the heavy-tailed structure of the count distribution becomes more pronounced as more data is included rather than averaged out.

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \text{ad}, \text{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair

Table 2: Plug-in delta-method asymptotic variance $\tilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\text{pairs}} = 4.61\text{M}$). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5 . The full- θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min [1]	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	4.53	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ -aware oracle	2.343	0.148	0.002	1.1

Table 3: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ tertile on full KDD (the bottom quartile boundary collapses into the lowest tertile because singletons dominate). Schemes are tied on the noise-dominated strata; harmonic dominates $\sim 5\times$ on the high-information stratum where the actual estimation signal lives.

$N_1 + N_2$ tertile	n pairs	$\tilde{V}_U$	ARE(h, u)
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	4.98

(1, 2) interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \leq 5$.

Headline (Table 2). ARE(harmonic, uniform) = 4.53: **uniform requires $\sim 4.5\times$ the impressions of harmonic for matching precision** on the relevance-agnostic surrogate. The min-count heuristic [1] achieves 4.41 on the same surrogate (-2.7% relative); harmonic and min are essentially tied empirically on KDD; harmonic’s contribution is the unconditional guarantee (Theorem 3) and the parameter-free derivation, not a large empirical margin. The count-only oracle (4.57) is the ceiling within the surrogate. The θ -aware oracle, which would be optimal under Aitken’s theorem with known r_i , plugs in $\hat{r}_i$ that is too noisy on the singleton-dominated tail and collapses to ARE = 0.002: a substantive finding rather than a side note—it shows that count-only weighting is robust precisely where plug-in IVW fails.

Asymmetric surrogate (empirical). With $\hat{p}_1 = 0.0501$, $\hat{p}_2 = 0.0256$ estimated from the marginal CTRs, $\alpha = 18.97$, $\beta = 38.08$, $\alpha/\beta \approx 0.50$. Harmonic reduces $\alpha V_k + \beta V_{k'}$ by a factor of 4.75 versus uniform. Both this number and the symmetric reduction of 4.84 are deterministic functionals of the observed counts—the gap reflects the intrinsic asymmetry of (α, β) , not Monte-Carlo noise—and the two are within 2% of each other. The closed-form asymmetric counterexample of Section 2 requires strongly *anti-correlated* $(N_k, N_{k'})$ across cells (each cell informative at exactly one position); on KDD the per-pair counts are positively correlated, as items popular at position 1 are typically also popular at position 2, and we conjecture this is generic for production logs.

Table 4: Count-only weighting schemes compared on full KDD via plug-in $\tilde{V}$. Schemes that concentrate more aggressively give lower $\tilde{V}$ but smaller ESS. Harmonic is the inverse-variance optimum among count-only schemes (Theorem 2).

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
$\log(1 + \min(N_k, N_{k'}))$	1.651	3.18e-3	1.71	3.17M
$\sqrt{N_k N_{k'}}$	1.706	2.28e-3	3.55	1,246
$\min(N_k, N_{k'})$	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	4.53	1,849
Harmonic, cap@99th pct	1.606	2.40e-3	2.83	765K

Stratified ARE (Table 3). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by $5\times$ on the high-information stratum. The gain comes from *not* treating singletons as equal to million-impression cells.

Weight-cap robustness. The harmonic ESS is 1,849 out of 4.61M pairs: concentration on a small subset of high-count cells is what produces the surrogate $\tilde{V}$ reduction, and the trade-off is sensitivity to those cells, which we probe directly by capping. Capping ω_i^h at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to 5.2×10^4 and still gives ARE = 4.12; at the 99th percentile, ARE = 2.83 (ESS 7.6×10^5); even at the 90th, ARE = 1.86. Harmonic beats uniform robustly across capping levels.

Comparison of count-only weighting schemes. Table 4 compares harmonic to the natural alternatives on full KDD. Among count-only weights, harmonic is best on $\tilde{V}$. Log-counting is too flat (treats singletons nearly as well as million-impression cells); the geometric mean and min-count concentrate more aggressively but neither matches the inverse-variance optimum. Capping at the 99th percentile preserves a $2.83\times$ advantage but shifts the estimand ($\hat{R} : 1.71 \rightarrow 1.61$), consistent with the estimand discussion below.

Estimand under PBM mis-specification. PBM is a known approximation; on real logs different weight schemes estimate different functionals of cell-specific ratios. Specifically, the population limit of $\hat{R}$ under any weight ω is

$$R_\omega = \frac{\sum_i \omega_i \theta_k^i}{\sum_i \omega_i \theta_{k'}^i},$$

which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is the same constant across cells—i.e. when PBM holds exactly. Under mis-specification each scheme thus targets a different weighted average of the cell-specific ratios $\rho_i = \theta_k^i/\theta_{k'}^i$, (uniform $\rightarrow 1.68$, harmonic $\rightarrow 1.71$, θ -aware oracle $\rightarrow 2.34$ on KDD). The plug-in $\tilde{V}$ comparison is a precision statement at fixed ω —a model-based efficiency comparison—and does *not* claim that one R_ω is the “correct” position-bias ratio. Choosing ω thus involves a precision/estimand trade-off; harmonic’s attractive property is that, among count-only schemes, it concentrates on high-information cells (where ρ_i is estimated most precisely) without depending on noisy plug-in $\hat{\theta}_i$.

Quantifying PBM mis-specification (Cochran’s Q). PBM is a known approximation, not a literal model – prior work [7, 15, 16] has long

Table 5: MSE decomposition under three explicit PBM violations (anchor imbalance 50:1, split 80/20, 5,000 trials). Count-only schemes share comparable bias² within each model; harmonic achieves the lowest variance, hence the lowest MSE.

Model	Scheme	MSE	Variance	Bias ²
Trust	Uniform	0.00487	0.00486	1.0e-5
	Min	0.00298	0.00297	2.0e-5
	Harmonic	0.00272	0.00272	0.0
Cascade	Uniform	0.01376	0.00276	0.01100
	Min	0.01242	0.00171	0.01071
	Harmonic	0.01252	0.00147	0.01105
Pos.-dep.	Uniform	0.04086	0.03973	0.00113
	Min	0.03896	0.03770	0.00127
	Harmonic	0.03724	0.03613	0.00111

established that real logs violate it. The relevant question is how *large* the deviation is. Cochran’s Q on the 16,132 high-information pairs (≥ 5 clicks at each of positions 1 and 2) gives $Q = 6.17 \times 10^4$, $df = 16,131$, $Q/df = 3.83$; per-cell odds-ratio variability is $\sim 3.8\times$ what binomial sampling alone would produce, well outside Poisson dispersion. This magnitude is the quantitative driver of the estimand drift discussed in Paragraph 4; plug-in $\tilde{V}$ keeps the efficiency comparison well-posed by working at fixed ω .

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1]: $M=10$ positions, $p_k = k^{-\eta}$ with $\eta = 1$, AdjacentChain estimator. A fraction s of documents are “anchors” with heavy traffic at one position ($N \sim \text{Unif}(50r, 200r)$) and light traffic at the other ($N \sim \text{Unif}(1, 51)$); the remaining $1 - s$ are balanced ($N \sim \text{Unif}(25, 75)$ at both positions); $s \in \{0.5, \dots, 0.95\}$, $r \in \{1, \dots, 100\}$, 500 trials per cell. At $s = 0.80$, $r = 100$ harmonic reduces variance by 43.6% vs. uniform and 5% vs. min; reduction grows to $\sim 54\%$ at $s = 0.90$, $r = 50$. Across the full (s, r) grid the data-equivalence factor $\text{DEF}(s, r) = b_{\text{uni}}(s, r)/b_{\text{harm}}(s, r)$ – the ratio of base impressions per document at which each scheme first hits a fixed MSE threshold – has median 1.31, mean 1.30, max 1.94. Highly imbalanced cells achieve up to $2\times$ data savings; near-balanced cells ($r = 1$) saturate at $\text{DEF} \approx 1$.

Bias comparability under PBM violations. We additionally simulate three explicit PBM violation models – trust bias, cascade, and position-dependent relevance [1, 6] – at imbalance ratio 50:1, split 80/20, 5,000 trials. Across all three models the count-only schemes (uniform, min, harmonic) exhibit *comparable* bias², so harmonic’s lower variance translates directly into lower MSE (Table 5). Under cascade, where history-dependent examination introduces a substantial structural mismatch (bias² ≈ 0.011 for every ratio-unbiased scheme), the variance edge is preserved but the practical MSE gain shrinks because bias dominates.

Yahoo-LTR semi-synthetic. Same simulator with graded relevance from the Yahoo Learning to Rank Challenge Set 1 test split (6,936

queries; relevance grades $g \in \{0, \dots, 4\}$ mapped to $r(q, d) = (2^g - 1)/15$ following NDCG; click model $P(C = 1 \mid q, d, k) = p_k r(q, d)$; 200 trials). Harmonic reduces variance by $\sim 40\%$ vs. uniform at imbalance $r = 50$ ($\text{Var } 1.5 \times 10^{-4} \rightarrow 8.7 \times 10^{-5}$). With 200 trials, a delta-method approximation places the 95% CI on the variance-ratio at $[0.49, 0.71]$, so the gain is well separated from 1.0 but not tightly pinned. Aggressive clipping trades variance for bias and yields $\sim 458\times$ higher MSE.

A production-style $K=5$ ranker A/B simulation (single hyperparameter setting) shows harmonic’s MSE reduction growing from 6.3% to 17.2% as adjacent-pair impression imbalance rises—qualitative direction only; full sweep deferred to the regular-paper version.

6 Related Work

Intervention harvesting was introduced by [1]; [15] jointly estimates relevance and propensities via EM. Self-normalised weighted ratio estimators are standard in importance sampling and survey statistics [4, 11, 12]; inverse-variance weighting traces to Aitken [2] and is canonical in meta-analysis [5, 8]. The closest off-policy analogue is SNIPS [3, 13]. PBM violations on real logs are documented in [7, 15, 16].

Why count-only PBM weighting is still worth studying. Given that PBM is approximate, EM-style joint estimation of relevance and propensities [15] is a natural alternative with its own bias-variance trade-offs (initialisation sensitivity, model choice). The two approaches are complementary: a relevance-agnostic weight is the right input to an EM M-step that re-uses count statistics. Empirical EM comparison is left to the regular-paper version.

7 Conclusion

Harmonic weighting is the inverse-variance optimum on a *relevance-agnostic count-only surrogate* of the PBM delta-method variance (Theorems 2–4). The actual inverse-variance solution (θ -aware oracle) is plug-in infeasible on singleton-dominated data ($\text{ARE} = 0.002$). The plug-in $\tilde{V}$ is the natural diagnostic at fixed ω ; on full KDD ARE(harmonic, uniform) = 4.53 on the surrogate, with harmonic and min essentially tied (4.53 vs. 4.41). Under known PBM misspecification (Cochran- $Q/df = 3.83$) different schemes target different weighted averages of cell-specific ratios, so choosing ω is a precision/estimand trade-off. Future work: shrinkage θ -aware oracle; EM comparison [15]; click models beyond PBM [7, 16].

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code, scripts, and data-processing pipelines are provided as supplementary material.

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